Lecture 1 (September 2nd)

Definition. (Contour integral) Let C be a contour parametrised by $z = z(t)$ where $a \leq t \leq b$. Also, let f be continuous on C. Then, the contour integral of f is defined as

$$\int_C f(z)\,dz = \int_C u\,dx - v\,dy + i\int_C u\,dy + v\,dx$$

and also, through parametrisation, we have

$$\int_C f(z)\,dz = \int_a^b f(z(t))z'(t)\,dt$$

Definition. (Derivative) The derivative of a complex function is defined as

$$f'(z_0) = \lim_{z\to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

if it exists. If f is C^1 (the function is continuous up its first derivatives) and $u_x = v_y$ & $v_x = -u_y$, then $f'(z)$ exists.

Theorem. (Green's theorem) Let P and Q be a C^1 function on $\bar{D}$ where D is a simply connnected bounded region whose boundary is C. Then,

$$\int_C P\,dx + Q\,dy = \iint_D \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big)\,dxdy$$

where C is oriented in the positive sense. Here,

$$\int_C P\,dx = -\iint_D \frac{\partial Q}{\partial y}\,dydx$$

and

$$\int_C Q\,dy = \iint_D \frac{\partial Q}{\partial x}\,dxdy$$

Proof. Notice that for a parametrised simple disc where the above is $y = g_2(x)$ and $y = g_1(x)$ on the bottom,

$$\int_C P\,dx = \int_a^b P(x, g_1(x))\,dx - \int_a^b P(x, g_2(x))\,dx$$

and

$$\iint_D \frac{\partial P}{\partial y}\,dydx = \int_a^b \int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y}\,dydx = \int_a^b \Big[P(x, g_2(x)) - P(x, g_1(x))\Big]\,dx$$

Similarly,

$$\int_D Q\,dy = \int_c^d Q(y, h_2(x))\,dy - \int_c^d Q(h_1(y), y)\,dy$$

and

$$\iint \frac{\partial Q}{\partial x}\,dxdy = \int_c^d \int_c^d \int_{h_1(y)}^{h_2(y)} \frac{\partial Q}{\partial x}\,dxdy = \Big[Q(h_2(y), y) - Q(h_1(y), y)\Big]\,dy$$

□

Theorem. (Cauchy's theorem) If f is a analytic (differentiable) inside and on a bounded simply connected domain D whose boundary is C and $f \in C^1(\bar{D})$ then

$$\int_C f(z)\,dz = 0$$

Proof. Let $f = u + iv$. Then $u_x = v_y$ and $v_x = -u_y$ on D. Thus

$$\int_C f(z)\,dz = \int_C u\,dx - v\,dy + i\Big[\int_C u\,dy + v\,dx\Big]$$

where

$$\int_C u\,dx - v\,dy = \iint_D \Big(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\Big)\,dxdy$$

and

$$\int_C u\,dy + v\,dx = \iint_D \Big(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\Big)\,dxdy$$

notice how the terms in the large brackets vanish, and that the integral itself equals zero. □

Theorem. (Generalised Cauchy's Theorem) If $f \in C^1(\bar{D})$ is analytic in a bounded multiply connected domain D with finitely many holes inside, then

$$\int_{\partial D} f(z)\,dz = 0$$

where ∂D is a positively oriented boundary of D.

Theorem. (Cauchy Integral Formula) If f is analytic inside and on a POSCC C then for every z_0 inside C we have

$$f(z_0) = \frac{1}{2\pi i}\int_C \frac{f(z)}{z - z_0}\,dz$$

Lecture 2 (September 4th)

Recall. (Cauchy's formula) For a positively oriented simple closed curve (POSCC) C and f that is analytic inside and on C, we have

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\xi)}{\xi - z}\, d\xi$$

for all z inside C. If $\bar{D}(z_0, r) = \{z \mid |z - z_0| \leq r\} \subset D$,

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0 + z_0 - z} = \frac{1}{(\xi - z_0)\left(1 - \dfrac{z - z_0}{\xi - z_0}\right)}$$

for $z \in D(z_0, r)$. However,

$$\left|\frac{z - z_0}{\xi - z_0}\right| < \frac{|z - z_0|}{r} < 1$$

so that

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0}\sum_{n=0}^{\infty}\left(\frac{z - z_0}{\xi - z_0}\right)^n = \sum_{n=0}^{\infty}\frac{(z - z_0)^n}{(\xi - z_0)^{n+1}}$$

Thus

$$\begin{aligned} f(z) =& \frac{1}{2\pi i}\int_C f(\xi)\sum_{n=0}^{\infty}\frac{(z - z_0)^n}{(\xi - z_0)^{n+1}}\, d\xi \\ =& \sum_{n=0}^{\infty} a_n (z - z_0)^n \end{aligned}$$

where

$$a_n = \frac{1}{2\pi i}\int_C \frac{f(\xi)}{(\xi - z_0)^{n+1}}\, d\xi = \frac{1}{n!} f^{(n)}(z_0)$$

Notice that the convergence is uniform for $\xi \in C$.

Theorem. If f_n is continuous in C such that $f_n \to f$ uniformly on C

$$\int_C f(z)\, dz = \lim_{n\to\infty}\int_C f_n(z)\, dz$$

Theorem. If $f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$ for $z \in D(z_0, R)$ then for $0 < r < R$ we have

$$\frac{1}{2\pi}\int_0^{2\pi} |f(z_0 + re^{it})|^2\, dt = \sum_{n=0}^{\infty} |c_n|^2 r^{2n}$$

Proof. Take

$$f(z_0 + re^{it}) = \sum_{n=0}^{\infty} c_n r^n e^{int}$$

so that

$$\begin{aligned}\frac{1}{2\pi}\int_0^{2\pi}|f(z_0+re^{it})|^2\,dt =& \frac{1}{2\pi}\int_0^{2\pi}\sum_{n=0}^{\infty}c_n r^n e^{int}\sum_{m=0}^{\infty}\bar{c}_m r^m e^{-imt}\,dt\\ &=\sum_{n=0}^{\infty}\sum_{m=0}^{\infty}\frac{1}{2\pi}\int_0^{2\pi}c_n\bar{c}_m r^{n+m}e^{i(n-m)t}\,dt\\ &=\sum_{n=0}^{\infty}|c_n|^2 r^{2n}\end{aligned}$$

Here, we note that

$$\int_0^{2\pi} e^{i(n-m)t} = \begin{cases} 2\pi & n=m \\ 0 & n\neq m\end{cases}$$

□

Corollary. (Liouville theorem) All bounded entire functions are constant. Using this theorem, we can prove the fundamental theorem of algebra.

Proof. If f is entire such that $|f(z)|\leq M$ for all $z\in \boldsymbol{C}$ then

$$\sum_{n=0}^{\infty}|c_n|^2 r^{2n}\leq M^2$$

for all $r>0$ which means that $c_1=c_2=\ldots=0$. □

Theorem. (Identity theorem) Let D be a domain (open and connected) and $\{z_n\}$ be a sequence of distinct points in D such that $z_n\to z_0\in D$. If f is analytic in D with $f(z_n)=0$ for all $n\in\boldsymbol{N}$, then $f(z)=0$ for all $z\in D$.

Theorem. (Maximum modulus theorem) Let f be analytic in a domain D. Then, $|f|$ can not have a local maximum in D unless it is constant.

Proof. If

$$|f(z_0+re^{it})\leq|f(z_0)|$$

for all $t\in[0,2\pi]$, then

$$\sum_{n=0}^{\infty}|c_n|^2 r^{2n}\leq|f(0)|^2=|c_0|^2$$

which means that $c_1=\ldots=c_n=0$. Then we can apply the identity theorem. □

Theorem. (Morera's theorem) Morera's theorem can be thought as Cauchy theorem's inverse. If f is continuous in a domain D such that

$$\int_C f(z)\,dz = 0$$

for all closed contours C in D, then f is analytic in D (note that this also means that the integral is path independent).

Proof. The key point is that for a given $a \in D$, $F(z) = \int_a^z f(\xi)\,d\xi$ is well defined. Take any $z_0 \in D$. We'll show that $F'(z_0) = f(z_0)$, which means that $f = F'$ is analytic. Take any $\varepsilon > 0$. We choose $\delta > 0$ so that

(i) $D(z_0, \delta) \subset D$

(ii) If $|z - z_0| < \delta$ then $|f(z) - f(z_0)| < \varepsilon$

We need to prove that if $0 < |z - z_0| < \delta$ then

$$\left|\frac{F(z) - F(z_0)}{z - z_0} - f(z_0)\right| < \varepsilon$$

Take the line segment path $[z_0, z]$ from z_0 to z. Then

$$F(z) - F(z_0) = \int_{[z_0,z]} f(\xi)\,d\xi$$

and

$$\frac{F(z) - F(z_0)}{z - z_0} - f(z_0) = \frac{1}{z - z_0}\int_{[z_0,z]} [f(\xi) - f(z_0)]\,d\xi$$

Notice that now, due to (ii), the term inside the square brackets can be reduced to less than ε since $|\xi - z_0| < \delta$. Therefore

$$\left|\frac{F(z) - F(z_0)}{z - z_0} - f(z_0)\right| < \frac{1}{|z - z_0|}|z - z_0|\varepsilon = \varepsilon$$

□

Theorem. Let $\{f_n\}$ be a sequence of analytic functions in a domain D such that $f_n \to f$ uniformly on every compact sbuset of D. Then f is analytic on D.

Example. On $\boldsymbol{R}$, consider

$$f_n(x) = \sum_{k=1}^{n} \frac{\cos 3^k x}{2^k} \to f(x) = \sum_{n=1}^{\infty} \frac{\cos 3^n x}{2^n}$$

where the convergence is uniform and the right-hand side is nowhere differentiable.

Proof. First, f is continuous on D since convergence is uniform. Let C be any closed contour in D. Then

$$\int_C f(z)\,dz = \lim_{n\to\infty} \int_C f_n(z)\,dz = 0$$

By Morera's theorem, the proof is complete. $\square$

Lecture 3 (September 9th)

Lemma. Let f be analytic on a simply connected domain D such that $f(z) \neq 0$ for any $z \in D$. Then,

(i) There is g analytic on D such that $e^g = f$ on D

(ii) There is h analytic on D such that $h^2 = f$ on D (this implies the Riemann mapping theorem)

Proof. (1) By Cauchy's theorem, the $\int_C f(z)\,dz = 0$ for every closed contour C in D. Thus for a given $z_0 \in D$, the integral

$$\int_{z_0}^{z} f(\xi)\,d\xi$$

is independent of path. Since f has no zero in D, f'/f is analytic on D. We define g on D by

$$g(z) = \int_{z_0}^{z} \frac{f'(\xi)}{f(\xi)}\,d\xi + \operatorname{Log} f(z_0)$$

Then we'll show that $f(z) = e^{g(z)}$ on D. Observe that

$$\begin{aligned}(fe^{-g})' =& f'e^{-g} + f(e^{-g})' \\ =& f'e^{-g} + fe^{-g}\Big(-\frac{f'}{f}\Big) \\ =& f'e^{-g} - f'e^{-g} = 0\end{aligned}$$

on D and $fe^{-g} = c$ a constant. If we define $h(z) = f(z)e^{-g(z)}$ then evaluating $h(z)$ at z_0 gives

$$\begin{aligned}h(z_0) =& f(z_0)e^{-g(z_0)} \\ =& f(z_0)e^{-\operatorname{Log} f(z_0)} \\ =& f(z_0)[f(z_0)]^{-1} = 1\end{aligned}$$

Therefore, $f(z) = e^{g(z)}$.

(2) By (1), there exists an analytic g such that $f = e^g$ on D. If we define $h = e^{g/2}$, then h is analytic on D with $h^2 = f$. $\square$

Theorem. (Exhastion of D by compact subsets) Let $D \subset \boldsymbol{C}$ be open. Then there is a sequence of compact subsets K_n of D such that

(i)
$$K_1 \subset K_2 \subset \ldots \subset K_n \subset D$$

(ii)
$$\bigcup_{n=1}^{\infty} K_n = D$$

(iii) If $K \subset D$ is compact then $K \subset K_n$ for some $n \in \boldsymbol{N}$

(iv) There is a sequence $\{\delta_n\}$ with $\delta_n > 0$ such that
$$\bar{D}(z, 2\delta_n) \subset K_{n+1}$$
for each $z \in K_n$

Proof. Define
$$K_n = \Big\{z \in \boldsymbol{C} \,\Big|\, |z| \leq n\Big\} \cap \Big\{z \in \boldsymbol{C} \,\Big|\, d(z, D^c) \geq \frac{1}{n}\Big\}$$
$$G_n = \Big\{|z| < n\Big\} \cap \Big\{d(z, D^c) > \frac{1}{n}\Big\}$$
and $G_n = D$ by the Archimeidan property. □

Lecture 4 (September 16th)

Theorem. Let D be a domain in $\boldsymbol{C}$. There is a sequence $\{K_n\}$ of compact subsets of D and $\{\delta_n\}$ ($\delta_n > 0$) such that

(i)
$$\bigcup_{n}^{\infty} K_n = D$$

(ii) If $K \subset D$ is compact then $K \subset K_n$ for some n

(iii) $\bar{D}(z, 2\delta_n) \subset K_{n+1}$ for every $z \in K_n$, $n = 1, 2, \ldots$

Note that $z \notin F$ and F is closed. This implies that $d(z, F) = \min\{|z - w| \,|\, w \in F\} > 0$.

Proof. Put
$$K_n = \Big\{z \leq n\Big\} \cap \Big\{d(z, D^C) \geq \frac{1}{n}\Big\}$$
□

Example. Consider

$$f_n(x) = \frac{\sin(nx)}{\sqrt{n}} \to 0$$

which converges uniformly on $\boldsymbol{R}$. However,

$$f'(0) = \sqrt{n} \to \infty$$

as $n \to \infty$.

Theorem. If $\{f_n\}$ is a sequence of analytic functions on a domain D such that $f_n \to f$ uniformly on every compact subset of D, then

(i) f is analytic on D (by Morera's theorem)

(ii) $f_n' \to f'$ uniformly on every compact subset of D

Proof. Take any compact subset K of D. Then $K \subset K_n$ for some n, and $\bar{D}(z, \delta_n) \subset K_{n+1}$ for all $z \in K$. So, there exists $r > 0$ and K', a compact subset of D, such that $\bar{D}(z, r) \subset K'$ for all $z \in K$. Here, applying Cauchy's integral formula,

$$f(z) = \frac{1}{2\pi i} \int_{|\xi - z| = r} \frac{f(\xi)}{(\xi - z)^2} \, d\xi$$

For every $z \in K$, we have

$$f'(z) = \frac{1}{2\pi i} \int_{|\xi - z| = r} \frac{f(\xi)}{(\xi - z)^2} \, d\xi$$

which implies that

$$|f'(z)| \leq \frac{1}{2\pi} \frac{||f||_{K'}}{r^2} (2\pi r) = \frac{||f||_{K'}}{r}$$

with $||f||_K = \max_{z \in K} |f(z)|$. Therefore, for every $z \in K$,

$$|f_n'(z) - f'(z)| \leq \frac{||f_n - f||_{K'}}{r}$$

wihch means $f_n' \to f'$ uniformly on K. $\square$

Example. (Linear mapping) Consider the linear mapping

$$w = az + b$$

This mapping we see that, by expressing $a = re^{i\theta}$, the map stretches, rotates, and translates the complex plane.

Definition. (Linear functional transformation) For $ad - bc \neq 0$,

$$w = T(z) = \frac{az+b}{cz+d}$$

This can be written as

$$w = T(z) = \frac{a}{c} + \frac{bc-ad}{c} \cdot \frac{1}{cz+d}$$

From this, we understand that the map is 1-1 as the following inverse map exists:

$$z = \frac{dw-b}{-cw+a}$$

Additionally, we see that the mapping can be seen as a combination of translation, dilation ($|a| > 1$) or contraction ($|a| \leq 1$), rotation, and inversion.

$$z \to \frac{1}{z}$$

This is the tricky one!

Remark. (Inversion map $w = 1/z$) The equation (locus) of all lines and all circles in $\boldsymbol{R}^2$ is

$$a(x^2 + y^2) + bx + cy + d = 0$$

where $b^2 + c^2 > 4ad$ and $a, b, c, d \in \boldsymbol{R}$. In the complex plane, for $x^2 + y^2 = |z|^2$, $x = (z + \bar{z})/2$ and $y = (z - \bar{z})/2i$ so that

$$a|z|^2 + \frac{1}{2}(b - ic)z + \frac{1}{2}(b + ic)\bar{z} + d = 0$$

and again $b^2 + c^2 > 4ad$. This implies that

$$a|z|^2 + \beta z + \bar{\beta}\bar{z} + d = 0$$

where $a, d \in \boldsymbol{R}$ and $\beta \in \boldsymbol{C}$ with $|\beta|^2 > ad$. Taking the inversion map,

$$a\frac{1}{|w|^2} + \beta\frac{1}{w} + \bar{\beta}\frac{1}{\bar{w}} + d|w|^2 = 0$$

and

$$a + \beta\bar{w} + \bar{\beta}w + d = 0$$

by multiplying $|w|^2 = w\bar{w}$ ($\beta\bar{\beta} > ad$). We therefore see that the inversion map takes the set of all lines and circles to again the set of all lines and circles.

Definition. (S^2) S^2 is defined as the boundary of a unit ball in $\boldsymbol{R}^3$. We see this as a one-point compactification (Alexandroff compactification) of $\boldsymbol{C}$, and is seen as $\boldsymbol{C} \cup \{\infty\}$ where $\{\infty\}$ is called the north pole.

Lemma. The linear functional transformation which takes $\{a, b, c\}$ to $\{0, 1, \infty\}$ is uniquely determined by

$$T(z) = \frac{(z-a)(b-c)}{(z-c)(b-a)}$$

Lecture 5 (September 18th)

Definition. Notice that the linear fractional transformation defined for $ad - bc \neq 0$ as

$$T(z) = \frac{az+b}{cz+d}$$

we see that the derivative is

$$T'(z) = \frac{ad-bc}{(cz+d)^2} \neq 0$$

and that it is analytic and T' is never 0. Meanwhile,

$$T^{-1}(z) = \frac{dz-b}{-cz+a}$$

Conformal are by definition, angle preserving maps (both size and orientation).

Definition. Let $z = z(t)$ for $a \leq t \leq b$ represent the contour C. Also, let $w = f(z)$, that is, $w(t) = f(z(t))$ represent the contour Γ. If $z_0 = z(t_0)$ and $w_0 = f(z_0) = f(z(t_0))$, f is analytic at z_0 such that $f'(z_0) \neq 0$. Then, from

$$w'(t_0) = f'(z_0)z'(t_0)$$

we have

$$\arg w'(t_0) = \arg f'(z_0) + \arg z'(t_0)$$

Thus, we see that analytic functions with nonzero deriviatives are conformal mappings.

Definition. Consider

$$T(z) = \frac{az+b}{cz+d}$$

(with $ad - bc \neq 0$) and

$$S(z) = \frac{ez+f}{gz+h}$$

(with $eh - gf \neq 0$) we ultimately have

$$(T \circ S)(z) = \frac{(ae+bg)z + (af+bh)}{(ce+dg)z + (cf+dh)}$$

also,

$$T^{-1}(z) = \frac{dz-b}{-cz+a}$$

Ultimately, we see that these mappings are equivalent to matrix multiplication.

Proposition. (The fixed points of $T(z) = z$) We see that

$$\frac{az+b}{cz+d} = z \qquad \Longrightarrow \qquad cz^2 + (d-a)z - b = 0$$

and that there can't be more than two fixed points. If S and T are linear fractional transformations such that $S(z_k) = T(z_k)$ for $k = 1, 2, 3$, then $S = T$ since $S^{-1} \circ T = I$.

Theorem. If distinct points $\{z_1, z_2, z_3\}$ and distinct points $\{w_1, w_2, w_3\}$ are points of $S^2 = \boldsymbol{C} \cup \{\infty\}$ then there exists a unique linear fractional transformation T such that $T(z_k) = w_k$ for $k = 1, 2, 3$.

$$\{a, b, c\} \to \{0, 1, \infty\} \leftrightarrow \{\alpha, \beta, \gamma\}$$

Then

$$T(z) = \frac{(b-c)(z-a)}{(b-a)(z-c)} \qquad \& \qquad S(z) = \frac{(\beta-\gamma)(z-\alpha)}{(\beta-\alpha)(z-\gamma)}$$

Example. Find V such that

$$\{0, 1, 2\} \to \{-1, 0, 4\}$$

Proof.

$$S(z) = \frac{-4(z+1)}{1(z-4)} = \frac{-4z-4}{z-4}$$

(we say it maps to$\{0, 1, \infty\}$) which is equivalent to

$$\begin{pmatrix} -4 & -4 \\ 1 & -4 \end{pmatrix}$$

For T that sends $\{0, 1, 2\}$ to the same points, we find the matrix, and $V(z)$ is given as

$$V(z) = \frac{8z-8}{-3z+8}$$

□

Example. Let

$$T(z) = \frac{1+z}{1-z}$$

which corresponds to

$$\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

Let $U = \{|z| < 1\}$, and say that we want to find $T(U)$. Notice that

$$\{-1, 0, 1\} \mapsto \{0, 1, \infty\}$$

As $T([-1, 1])$ maps to the positive x-axis, we can see that $T(-1) = \infty$ and that $T(|z| = 1)$ is a line passing $T(-1) = 0$. Then, $T(i) = i$ and $T(|z| = 1)$ is the y-axis. Another way to do this is through conformality of $T(|z| = 1)$ and the positive real axis being $\pi/2$ at $z = 0$.

Definition. Let U be the unit disk and $T = \{|z| = 1\}$ be a unit circle. For $\alpha \in U$, we define

$$\phi_\alpha(z) = \frac{z - \alpha}{1 - \bar{\alpha}z} : \begin{pmatrix} 1 & -\alpha \\ -\bar{\alpha} & 1 \end{pmatrix}$$

Then ϕ_α is analytic on $\bar{U}$ such that $\phi_\alpha^{-1} = \phi_{-\alpha}$, $\phi_\alpha(0) = -\alpha$, $\phi_\alpha(\alpha) = 0$, $\phi_\alpha(U) = U$, $\phi_\alpha(T) = T$, and $\phi'_\alpha(0) = 1 - |\alpha|^2$, $\phi'(\alpha) = 1/(1 - |\alpha|^2)$.

Proof. We'll show that $\phi_\alpha(T) = T$.

$$\begin{aligned} \phi_\alpha(e^{it}) =& \frac{e^{it} - \alpha}{1 - e^{-it}\bar{\alpha}} = \frac{e^{it}(1 - e^{-it}\alpha)}{1 - e^{-it}\bar{\alpha}} \\ |\phi_\alpha(e^{it})| = 1 & \end{aligned}$$

□

Lecture 6 (September 23rd)

Recall. Last class, we've took $U = \{|z| < 1\}$ and $T = \{|z| = 1\}$ with $\alpha \in U$. The map $\phi_\alpha(z)$ was defined as

$$\phi_\alpha(z) = \frac{z - \alpha}{1 - \bar{\alpha}z} : \begin{pmatrix} 1 & -\alpha \\ -\bar{\alpha} & 1 \end{pmatrix}$$

We've noted that $1 - \bar{\alpha}z = 0$ implied that $z = \overline{(1/\alpha)} \notin \bar{U}$. You see that ϕ_α is analytic and one-to-one. We find explicitly that

$$\phi_\alpha^{-1} = \begin{pmatrix} 1 & \alpha \\ \bar{\alpha} & 1 \end{pmatrix}$$

such that $\phi_\alpha^{-1} = \phi_{-\alpha}$

$$\phi_\alpha^{-1}(z) = \phi_{-\alpha} = \frac{z + \alpha}{1 + \bar{\alpha}z}$$

and

$$\phi'_\alpha(z) = \frac{1 - |\alpha|^2}{(1 - \bar{\alpha}z)^2}$$

so that $\phi'_\alpha(0) = 1 - |\alpha|^2$ and $\phi'_\alpha(\alpha) = 1/(1-|\alpha|^2)$. What about mappings from T? If $z = e^{it}$, then

$$\phi_\alpha(e^{it}) = \frac{e^{it} - \alpha}{1 - \bar{\alpha}e^{it}} = \frac{e^{it} - \alpha}{e^{it}(e^{-it} - \bar{\alpha})} = \frac{1}{e^{it}} \frac{e^{it} - \alpha}{\overline{(e^{it} - \alpha)}}$$

such that $|\phi_\alpha(e^{it})| = 1$ and $\phi_\alpha(T) \subset T$. Furthermore, we know that $\phi_{-\alpha}(T) \subset T$. We thus find, that $\phi_\alpha(T) = T$. With this information, via the maximum modulus theorem, we find that

$$\max_{z \in \bar{U}} |\phi_\alpha(z)| = \max_{z \in T} |\phi_\alpha(z)| = 1$$

which implies that

$$|\phi_\alpha(z)| < 1$$

for all $z \in U$.

Remark. We summerize like the following:

(i) For $\alpha \in U$,
$$\phi_\alpha(z) = \frac{z - \alpha}{1 - \bar{\alpha}z}$$

(ii) $\phi_\alpha : U \to U$ is one-to-one, analytic and onto.

(iii) $\phi_\alpha(T) = T$

(iv) $\phi'_\alpha(0) = 1 - |\alpha|^2$

(v) $\phi'_\alpha(\alpha) = 1/(1 - |\alpha|^2)$

Theorem. (Schwartz lemma) Let $f : U \to U$ is analytic with $f(0) = 0$. For every $z \in U$, we have $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$. Furthermore, if one of the equalities hold, for some $z \in U\backslash\{0\}$ then f is a rotation.

$$f(z) = e^{i\theta} z$$

for some θ.

Proof. Notice that there exists a g analytic on U such that $f(z) = zg(z)$. For every fixed $z \in U$,

$$\begin{aligned} |g(z)| = \left|\frac{f(z)}{z}\right| &\leq \max_\theta \left|\frac{f(re^{i\theta})}{r}\right| \\ &\leq \frac{1}{r} \end{aligned}$$

if $|z| < r < 1$. Taking $r \to 1^-$, $|f(z)/z| \leq 1$. That is, $|f(z)| \leq |z|$ for all $z \in U$. Also, $f'(z) = g(z) + zg'(z)$ which implies that $f'(0) = g(0)$. This again implies that $|f'(0)| = |g(0)| \leq 1$. Notice how if the first equality holds, $|f(z)| = |z|$, and the function f only rotates the input. If the second equality holds, $|f'(0)| \leq 1$, and $|f(z)| \leq |z|$, meaning the same thing!

Notice the following. If $f(0) = \alpha$, then define $g(z) = \phi_\alpha(f(z))$. Then, $g : U \to U$ is analytic with $g(0) = 0$. Applying the Schwartz lemma to g, $|g'(0)| \le 1$ implies that

$$|f'(0)| \le 1 - |\alpha|^2$$

But,

$$g'(0) = \phi'_\alpha(f(0))f'(0) = \phi'_\alpha(\alpha)f'(0) = \frac{1}{1 - |\alpha|^2} f'(0)$$

□

Theorem. (Pick theorem) If $f : U \to U$ is analytic with $f(\alpha) = \beta$, then

$$|f'(\alpha) \le \frac{1 - |\beta|^2}{1 - |\alpha|^2}$$

The equality holds when

$$f(z) = \phi_{-\beta}(e^{i\theta}\phi_\alpha(z))$$

Proof. We define g on U so that $g(0) = 0$. Indeed,

$$g = \phi_\beta \circ f \circ \phi_{-\alpha}$$

is analytic from U into U with $g(0) = 0$. Applying the Schwartz lemma to g, we have $|g'(0)| \le 1$. But

$$g'(0) = \Big(\phi'_\beta[f(\phi_\alpha(0))]\Big) f'(\phi_\alpha(0))\phi'_\alpha(0) = \phi'_\beta(\beta) \cdot f'(0) \cdot \phi'_{-\alpha}(0) = \frac{1 - |\alpha|^2}{1 - |\beta|^2} f'(0) \le 1$$

When does the equality hold? Let $g(z) = \lambda z$ for some $|\lambda| = 1$. Then

$$\phi_\beta(f(\phi_{-\alpha}(z))) = \lambda z$$

By putting $z \to \phi_\alpha(z)$,

$$\phi_\beta(f(z)) = \lambda\phi_\alpha(z)$$

and

$$\phi_{-\beta}\Big[\phi_\beta(f(z))\Big] = \phi_{-\beta}(\lambda\phi_\alpha(z))$$

and ultimately

$$f(z) = \phi_{-\beta}(\lambda\phi_\alpha(z))$$

□

Theorem. If f is one-to-one analytic from U onto U, then there exists $\alpha \in U$ and $\theta \in \boldsymbol{R}$ such that $f(z) = e^{i\theta}\phi_\alpha(z)$ for all $z \in U$.

Lecture 7 (September 25th)

Theorem. (Schwartz-Pick theorem) Let $f : U \to U$ be an analytic function such that $f(\alpha) = \beta$. Then,

$$|f'(\alpha)| \leq \frac{1 - |\beta|^2}{1 - |\alpha|}$$

and the equality holds when

$$f(z) = \phi_{-\beta}(\lambda \phi_\alpha(z))$$

for some $\lambda \in T$.

Example. Observe that $f(x) = x^3$ is one-to-one on $\boldsymbol{R}$, but $f'(0) = 0$. Therefore, we see that on $\boldsymbol{R}$, f being one-to-one on an interval I doesn't necessarily imply that $f'(x) = 0$ for any $x \in I$. However, the converse works as if $f'(x) \neq 0$ for any $x \in I$, then f is one-to-one by the mean value theorem.

However, on $\boldsymbol{C}$, $f(z) = e^z$ satisfies $f'(z) \neq 0$ for any $z \in \boldsymbol{C}$ but f is not one-to-one. We claim that if f is analytic and one-to-one on a domain D, then $f'(z) \neq 0$ for any $z \in D$.

Theorem. If $f : U \to U$ is one-to-on analytic, meaning that $f'(z) \neq 0$ for any $z \in U$, then $f(z) = e^{i\theta}\phi_\alpha(z)$ for some $\alpha \in U$, $\theta \in \boldsymbol{R}$

Proof. Let $f(\alpha) = 0$. First define $g = f^{-1}$ on U. Then $g(f(z)) = z$ for all $z \in U$ and g is analytic on U. Due to the fact that the derivative is never zero, we can use the inverse function theorem, where we are able to write

$$\frac{g(w) - g(w_0)}{w - w_0} = \frac{z - z_0}{f(z) - f(z_0)} = \frac{1}{\frac{f(z) - f(z_0)}{z - z_0}}$$

and

$$g'(w_0) = \frac{1}{f'(z_0)}$$

since $f'(z_0) \neq 0$. We see that

$$g'(0)f'(\alpha) = 1$$

by the chain rule. By the Schwartz-Pick theorem,

$$|f'(\alpha)| \leq \frac{1}{1 - |\alpha|^2}$$

and $|g'(0)| \leq 1 - |\alpha|^2$. This means that

$$\frac{1}{1 - |\alpha|^2} \leq |f'(\alpha)| \leq \frac{1}{1 - |\alpha|^2}$$

By the Schwartz lemma,

$$f(z) = \lambda \phi_\alpha(z)$$

for some $\lambda \in T$. $\square$

Theorem. If f is analytic in a domain D with $f'(z_0) \neq 0$ for some $z_0 \in D$ then there is $r > 0$ such that $D(z_0, r) \subset D$ and f is one-to-one on $D(z_0, r)$.

Proof. If $z, w \in D(z_0, r)$ and $z \neq w$, then

$$\Big| \frac{f(z) - f(w)}{z - w} - f'(z_0) \Big| < \frac{1}{2} |f'(z_0)|$$

which implies that

$$\frac{1}{2} |f'(z_0)| < \Big| \frac{f(z) - f(w)}{z - w} \Big|$$

if $z, w \in D(z_0, r)$ with $z \neq w$. $\square$

Lemma. If f is analytic in a domain D and

$$g(z, w) = \begin{cases} \dfrac{f(z) - f(w)}{z - w} & z \neq w \\ f'(z) & z = w \end{cases}$$

then g is continuous on $D \times D$. Note that this is trivially true on $\boldsymbol{R}$.

Lemma. If $|b - a| < \dfrac{|a|}{2}$, then $|b| > \dfrac{|a|}{2}$.

Proof. Notice that if suffices to prove the statement only when $z = w$. Since f' is continuous on D, for every ε, there exists a δ such that if $z \in D(a, \delta)$ then $|f'(z) - f'(a)| < \varepsilon$. Take any z, w ($z \neq w$) in $D(a, r)$. Define $\xi(t) = tz + (1 - t)w$ for $0 \leq t \leq 1$. Then,

$$\int_0^1 f'(\xi(t))\, dt = \frac{f(z) - f(w)}{z - w} = g(z, w)$$

so that

$$g(z, w) - g(a, a) = \int_0^1 [f'(\xi(t)) - f'(a)]\, dt$$

Since $z, w \in D(a, r)$ we have

$$\Big| \int_0^1 [f'(\xi(t)) - f'(a)]\, dt \Big| \leq \int_0^1 |f'(\xi(t)) - f'(a)|\, dt < \varepsilon$$

In conclusion, if $z, w \in D(a, r)$, then $|g(z, w) - g(a, a)| < \varepsilon$. $\square$

Theorem. (Rouche's theorem) If f and g are analytic inside and on a simple, closed contour C such that $|g(z)| < |f(z)|$ for every $z \in C$, then f and $f+g$ have the same number of zeros inside C.

Theorem. (Open mapping theorem) Let f be non-constant analytic in a domain Ω. If $D \subset \Omega$ is open, then $f(D)$ is open.

Proof. To show that $f(D)$ is open, take any $w_0 \in f(D)$. We need to find some $m > 0$ such that $D(w_0, m) \subset f(D)$. Let $f(z_0) = w_0$. Then there is $r > 0$ such that

$$\bar{D}(z_0, r) \subset D$$

and $f(z) - w_0$ has no zeros in $0 < |z - z_0| \leq r$ by the identity theorem. Let

$$m = \min_{|z-z_0|=r} |f(z) - w_0| > 0$$

Then we'll show that $D(w_0, m) \subset f(D)$. Take any $w_1 \in D(w_0, m)$ (we're trying to show that $w_1 \in f(D)$). Then

$$|w_0 - w_1| < |f(z) - w_0|$$

on $|z - z_0| = r$. Rouche's theorem implies that $f(z) - w_0$ and $f(z) - w_1$ have the same number of zeros inside $|z - z_0| = r$ which implise that

$$w_1 \in f(D(z_0, r)) \subset f(D)$$

□

Theorem. If f is one-to-one analytic in a domain D, then $f'(z) \neq 0$ for any $z \in D$.

Proof. Suppose that $f'(z_0) = 0$ for some $z_0 \in D$ (we'll show that f is not one-to-one). Then $f(z) - f(z_0) = f''(z_0)(z - z_0)^2 + \dots$ in some neighborhood of z_0. There is $r > 0$ such that $\bar{D}(z_0, r) \subset D$ and $f(z) - f(z_0)$ and $f'(z)$ both have no zeros in $0 < |z - z_0| \leq r$.

Let $m = \min_{|z-z_0|=r} |f(z) - f(z_0)|$ and choose any $w_1 \in D(f(z_0), m)$. Then,

$$|f(z_0) - w_1| < m \leq |f(z) - f(z_0)|$$

on $|z - z_0| = r$. By Rouche's theorem, $f(z) - f(z_0)$ and $f(z) - w_1$ have the same number of zeros in $D(z_0, r)$ where $f(z) - f(z_0)$ has at least 2 zeros. In other words, since $f' \neq 0$ in $D(z_0, r)$, there is at least 2 distinct zeros of $f(z) - w_1$ in $D(z_0, r)$ where f is not one-to-one. □

Lecture 8 (September 30th)

Theorem. Let $\{f_n\}$ be a sequence of 1-1 analytic functions on a domain D such that $f_n \to f$ uniformly on every compact subset of D. Then, either f is constant or f is 1-1 on D.

Proof. Let f be non-constant. Suppose that there exists $z_1, z_2 \in D$ such that $f(z_1) = f(z_2) = a$ and that the function is not 1-1. Then, there exists $r > 0$ such that:

(i) $\bar{D}(z_1, r) \subset D$ and $\bar{D}(z_2, r) \subset D$

(ii) By the identity theorem, $f(z) - a$ has no zeros in both $0 < |z - z_1| \leq r$ and $0 < |z - z_2| \leq r$. Denote simple circles $C_1 : |z - z_1| = r$ and $C_2 : |z - z_2| = r$. Let

$$m = \min_{z \in C_1 \cup C_2} |f(z) - a| \geq 0$$

where $C_1 \cup C_2$ is compact.

(iii) $\bar{D}(z_1, r) \cap \bar{D}(z_2, r) = \varnothing$

By the uniform convergence of $\{f_n\}$ on $C_1 \cup C_2$, there is N such that

$$|f_N(z) - f(z)| < m$$

for $z \in C_1 \cup C_2$. On C_1, $|f_N(z) - f(z)| < m \leq |f(z) - a|$. By Rouche's theorem, $f(z) - a$ and $f_N(z) - a$ have the same numbers of zero inside C_1. Also, $f(z) - a$ and $f_N(z) - a$ have the same number of zeros inside C_2. Thus, $f_N(z) - a$ has zeros inside C_1 and inside C_2. Thus, f_N is not 1-1. □

Example. There are multiple cases where the first option holds. For example, take $f_n(z) = \dfrac{z}{n}$ and $f = 0$.

Theorem. (Riemann mapping theorem) If D ($\neq \boldsymbol{C}$) is a simply connected domain, then there is a 1-1 analytic function h from D onto U.

Proof. Let $\Sigma = \{f : D \to U \mid f \text{ is injective and analytic}\}$.

(I) We prove that $\Sigma \neq \varnothing$. To prove this, we use the prior result that if f is analytic on a simply connected domain D, such that $f(z) \neq 0$ for any $z \in D$, then there is g analytic in D such that $g^2 = f$ on D. Suppose that $w_0 \notin D$. Then there is g analytic on D such that $g^2(z) = z - w_0$. As $w^2 = z$ implies that $w = \sqrt{r}e^{i\theta/2}$, $g(D)$ does not intersect some half of the plane! Furthermore,

(i) $0 \notin g(D)$, which is trivial.

(ii) g is 1-1 on D, as if $g(z_1) = g(z_2)$, then $g^2(z_1) = g^2(z_2)$ and $z_1 - w_0 = z_2 - w_0$.

(iii) If $w \in g(D)$, then $-w \notin g(D)$, as if $g(z_1) = -g(z_2)$ for distinct points, then $g^2(z_1) = g(z_2)^2$ and $z_1 = z_2$ which is a contradiction.

(iv) If $a \in g(D)$, then there exists an $r > 0$ such that $D(a, r) \subset g(D)$ due to the open mapping theorem. In particular, $D(-a, r) \cap g(D) =$ as if $w \in D(-a, r)$, then $|w+a| < r$ and $|-w-a| = |w+a| < r$ which implies that $-w \in D(a, r) \subset g(D)$.

Now, define $h : D \to \boldsymbol{C}$ by

$$h(z) = \frac{r}{g(z) + a}$$

Notice that $|g(z) + a| > r$ for all $z \in D$ and that $|h(z)| < 1$ for all $z \in D$. We see that if $h(z_1) = h(z_2)$.

$$\frac{r}{g(z_1) + a} = \frac{r}{g(z_2) + a}$$

and that $g(z_1) = g(z_2)$ with $z_1 = z_2$. We see that $h(z)$ is a 1-1 function from an arbitrary domain D to U!

(II) Let $\psi \in \Sigma$ such that $\psi(D) \neq U$. We claim that for every $z_0 \in D$, there is $\psi_0 \in \Sigma$ such that $|\psi_0'(z_0)| > |\psi'(z_0)|$. Suppose that $\alpha \in U \ \psi(D)$. Then, $\phi_\alpha \circ \psi$ is analytic on D with no zeros. Since D is simply connected, there is g analytic in D such that $g^2 = \phi_\alpha \circ \psi$. Let $g(z_0) = \beta$. Define $\psi_0 = \phi_\beta \circ g$. Then, $\psi_0 \in \Sigma$ and due to the Schwartz lemma, the claim holds.k $\square$

Lecture 9 (October 2nd)

Proof. (I) Take any $w_0 \notin D$, and then there is g that is analytic in D such that $g^2(z) = z - w_0$. If $D(a, r) \subset g(D)$, then $D(-a, r) \cap g(D) = \varnothing$. Now define

$$h(z) = \frac{r}{g(z) + a}$$

then $h \in \Sigma$.

(II) We claim that if $\psi \in \Sigma$ and $\psi(D) \neq U$, then for every $z_0 \in D$, these is $\psi_0 \in \Sigma$ such that $|\psi_0'(z_0)| > |\psi'(z_0)|$. Let $\alpha \neq \psi(D)$, then $\phi_\alpha \circ \psi$ is analytic in D with no zeros in D. Thus, there is g analytic in D such that $g^2 = \phi_\alpha \circ \psi$ in D. Given $z_0 \in D$, if $\beta = g(z_0)$, then $\psi_0 = \phi_\beta \circ g$ satisfies the claim.

Notice how $g \in \Sigma$ since g is 1-1 in D (if $g(z_1) = g(z_2)$, then $(\phi_\alpha \circ \psi)(z_1) = (\phi_\alpha \circ \psi)(z_2)$ concluding that $z_1 = z_2$). Additionally denote the square function by $s(z) = z^2$. Then,

$$\phi_\alpha \circ \psi = g^2 = s \circ g$$

so that $\psi = \phi_{-\alpha} \circ s \circ g$. Additionally, $\psi_0 = \phi_\beta \circ g$ so that $g = \phi_{-\beta} \circ \psi_0$. Therefore,

$$\psi = \phi_{-\alpha} \circ s \circ \phi_{-\beta} \circ \psi_0$$

on D. Now we will show that the claimed inequality holds. Defining $F = \phi_\alpha \circ s \circ \phi_{-\beta}$ or $F : U \to U$ is analytic. By the Schwartz lemma,

$$|F'(0)| \leq 1$$

since F is not 1-1. From $\psi = F \circ \psi_0$, we have

$$\psi'(z_0) = F'(\psi_0(z_0))\psi_0'(z_0) = F'(0)\psi_0'(z_0)$$

Since ψ is 1-1, the derivative $\psi' \neq 0$ in D and we have proved the claim.

(III) We now claim that if we take any $z_0 \in D$ and let $\eta_0 = \sup\{|\psi'(z_0)| \,|\, \psi \in \Sigma\}$, there is $h \in \Sigma$ such that $|h'(z_0)| = \eta_0$. From this, as $h(D)$ doesn't conform to the condition of step (II), we will have that $h(D) = U$. The key method here is using "normal families". $\square$

Remark. On $\boldsymbol{R}^n$,

(i) If G is a bounded infinite, then G has a limit point

(ii) If $\{a_n\}$ is a bounded sequence, then $\{a_n\}$ has a convergent subsequence

(iii) If K is closed and bounded, then every open cover of K has a finite subcover

However, consider the following. The Hilbert space $l^2(\boldsymbol{N})$ can be seen as the space closest to R^∞.

$$\mathbf{x} = (x_1, x_2, \ldots) \subset l^2(\boldsymbol{N}) \qquad \text{if} \qquad ||\mathbf{x}||^2 = \sum_{n=1}^{\infty} x_n^2 < \infty$$

Defined this manner, $l^2(\boldsymbol{N})$ is a complete normed vector space. Consider $E = \{\mathbf{e}_n \,|\, n \in \boldsymbol{N}\} \subset l^2(\boldsymbol{N})$ where $\mathbf{e}_k = (0, \ldots, 0, 1, 0, \ldots)$ where only the k-th element is nonzero. This basis satisfies $||\mathbf{e}_k|| = 1$.

(i) Then, E is a bounded infinite subset of $l^2(\boldsymbol{N})$ which has no limit point.

(ii) Also, E is a bounded sequence in $l^2(\boldsymbol{N})$ with no convergent subsequence.

(iii) Lastly, E is closed and bounded in $l^2(\boldsymbol{N})$ but is not compact.

Definition. (Three types of convergences & boundednesses) If D is open and $f_n : D \to \boldsymbol{C}$, then we can define three types of convergences.

(i) Pointwise convergence in D

(ii) Uniform convergence in D

(iii) Uniform convergence in every compact subset of D

There are also three types of boundednesses of $\{f_n\}$ on D

(i) Uniformly bounded on D, meaning that $|f_n| \leq M$ for all $n \in \boldsymbol{N}$ and $z \in D$.

(ii) Pointwise bounded on D, meaning there is $M_z > 0$ such that $|f_n(z)| \leq M_z$ for every $n \in \boldsymbol{N}$.

(iii) Uniformly bounded on every compact subset of D, meaning that for every compact subset of K of D, there is $M_K > 0$ such that $|f(z)| \leq M_K$ for all $n \in \boldsymbol{N}$ and $z \in K$.

Theorem. (Arzela-Ascoli theorem) A separable metric space is space that has a countable dense subset and a metric space is a space endowed with a metric. If $\{f_n\}$ is a pointwise bounded and equicontinuous sequence of functions on a separable metric space, $\{f_n\}$ has a subsequence that converges uniformly on all compact subsets.

Lecture 10 (October 14th)

Theorem. (Arzerla Ascoli) Let $\{f_n\}$ be a pointwise bounded and equicontinuous sequence of functions on a separable metric space X. Then, $\{f_n\}$ has a subsequence which converges uniformly on all compact subsets of X. This subsequence converges pointwise since every $\{x_n\} \subset X$ is compact.

Definition. (Pointwise bounded) A sequence $\{f_n\}$ is pointwise bounded on X provided that for every $x \in X$ there is $M_x > 0$ such that $|f_n(x)| \leq M_x$ for all $n \in \boldsymbol{N}$.

Definition. (Equicontinuity) A sequence $\{f_n\}$ is equicontinuous on X provided that for every $\varepsilon > 0$, there is $\delta > 0$ such that if $d(x, y) < \delta$ then $|f_n(x) - f_n(y)| < \varepsilon$ for all $n \in \boldsymbol{N}$.

Remark. If $\{f_n\}$ is equicontinuous on X, then each f_n is uniformly continuous on X. The converse is not true, for example set $f_n : \boldsymbol{R} \to \boldsymbol{R}$ as

$$f_n(x) = nx$$

Notice how $|f_n(x) - f_n(y)| = n|x - y|$ and each f_n is uniformly continuous, but $\{f_n\}$ is not equicontinuous on $\boldsymbol{R}$.

Proof. Let $E = \{x_1, x_2, \ldots\}$ be a countable dense subset of X. The proof is done in two steps:

(Step 1) We need to use pointwise boundedness to find a subsequence $\{g_n\}$ of $\{f_n\}$ which converges pointwise on E. Notice that $\{f_n(x_1)\}$ is a bounded sequence in $\boldsymbol{C}$, such that it has a convergent subsequence $\{f_{n,1}(x_1)\}$. Then, $\{f_{n,1}(x_2)\}$ is bounded and has a bounded subsequence in $\boldsymbol{C}$, such that it has a convergent subsequence $\{f_{n,2}(x_2)\}$. Notice that meanwhile, $\{f_{n,2}(x_1)\}$ also converges. Continuing and taking the diagonal, we have $\{f_{n,n}(x)\}$ which converges for every $x_n \in E$. Set $g_n = f_{n,n}$, and we see that for any x_m, $g_m, g_{m+1}, \ldots$ is a subsequence of $\{f_{n,m}\}_{n=1}^{\infty}$ which converges at x_m.

(Step 2) We now use equicontinuity to prove that this sequence converges uniformly on any compact subset K of X. We now use equicontinuity to show that there exists a $N \in \boldsymbol{N}$ satisfying if $n, m > N$ then $|f_n(x) - f_m(x)| < \varepsilon$ for every $x \in K$ ($\{f_n\}$ is uniformly Cauchy on K). Observe that due to the equicontinuity of $\{g_n\}$, for every $\varepsilon > 0$, there exists $\delta > 0$ such that if $p, q \in X$ satisfies $d(p,q) < \delta$, then $|f_n(p) - f_n(q)| < \varepsilon/3$. Also,

$$\mathcal{F} = \left\{ B\left(x, \frac{\delta}{2}\right) \;\middle|\; x \in K \right\}$$

is an open cover of K so that there is a finite subcover $\{B_1, B_2, \ldots, B_M\}$ (here, note that if $x, y \in B_k$, then $d(x,y) < \delta$). Since E is dense in X, then $p_i \in E \cap B_i$ for $1 \le i \le M$. From step 1, $\lim_{n\to\infty} g_n(p_i)$ converges for $1 \le i \le M$ and thus $\{g_n(p_i)\}$ is a Cauchy sequence for $1 \le i \le M$. Thus there is $N \in \boldsymbol{N}$ such that if $n, m > N$ then $|g_n(p_i) - g_m(p_i)| < \varepsilon/3$ for $1 \le i \le M$ with $N = \max\{N_1, \ldots, N_M\}$. Now we show the same for an arbitrary $x \in K$.

$$\begin{aligned}
|g_n(x) - g_m(x)| &= |g_n(x) - g_n(p_k) + g_n(p_k) - g_m(p_k) + g_m(p_k) - g_m(x)| \\
&\le |g_n(x) - g_n(p_k)| + |g_n(p_k) - g_m(p_k)| + |g_m(p_k) - g_m(x)| \\
&\le \varepsilon
\end{aligned}$$

as $d(x, p_k), d(x, p_m) < \delta$. □

Definition. (Uniformly bounded on a compact K) For every compact $K \subset D$, there is $M_K > 0$ such that $|f_n(z)| \le M_K$ for all $z \in K$, $m \in \boldsymbol{N}$.

Theorem. (Montel's theorem) If $\{f_n\}$ is a squence of analytic functions on a domain D which is uniformly bounded on each compact subset of D, then $\{f_n\}$ has a subsequence which converges uniformly on all compact subsets of D.

Theorem. If $\{f_n\}$ is a sequence of analytic functions on a domain D which is uniformly bounded on a compact subset of D, then $\{f_n\}$ is equicontinuous on every compact subset of D.

Proof. The proof is done through the Cauchy integral formula along with the exhaustion of D by compact subsets. □

Lecture 11 (October 16th)

Theorem. Let $\{f_n\}$ be analytic in D uniformly bounded on each compact subset. Then, $\{f_n\}$ is equicontinuous on every compact subset of D.

Proof. Take any compact subset K of D. We'll show that $\{f_n\}$ is equicontinuous on K. Take any $\varepsilon > 0$, and we have to find $\delta > 0$ such that if $z, w \in K$ satisfies $|z - w| < \delta$, then $|f_n(z) - f_n(w)| < \varepsilon$ for all $n \in \boldsymbol{N}$.

By the exhausion of compact subsets, there is a sequence of compact sets $\{K_n\}$ and $\delta_n > 0$ such that $\bigcup_{n=1}^{\infty} K_n = D$, and for every $z \in K_n$ we have $\bar{D}(z, 2\delta_n) \subset K_{n+1}$ for each n. Take $N \in \boldsymbol{N}$ such that $K \subset K_N$ so that for $z, w \in K$ and $|z - w| < \delta_N$ we have

$$f_n(z) = \frac{1}{2\pi i} \int_{|\xi - z| = 2\delta_N} \frac{f_n(\xi)}{\xi - z} \, d\xi \qquad \& \qquad f_n(w) = \frac{1}{2\pi i} \int_{|\xi - z| = 2\delta_N} \frac{f_n(\xi)}{\xi - w} \, d\xi$$

(where w is inside $|\xi - z| = 2\delta_N$) so that for $z, w \in K$ with $|z - w| < \delta_N$,

$$f_n(z) - f(w) = \frac{z - w}{2\pi i} \int_{|\xi - z| = 2\delta_N} \frac{f_n(\xi)}{(\xi - z)(\xi - w)} \, d\xi$$

Noting that $|f_n(\xi)| \leq M_{K_{N+1}}$ and $|\xi - z| = 2\delta_N$ implying $|\xi - w| > \delta_N$, we have

$$|f_n(z) - f_n(w)| \leq \frac{|z - w|}{2\pi} \frac{M_{K_{N+1}}}{(2\delta_N)\delta_N} \cdot 4\pi\delta_N = \frac{M_{K_{N+1}}}{\delta_N} |z - w|$$

So far we have proved that if $z, w \in K$ satisfies $|z - w| < \delta_N$, then

$$|f_n(z) - f_n(w)| \leq \frac{M_{N+1}}{\delta_N} |z - w|$$

for all $n \in \boldsymbol{N}$. Take

$$\delta_n = \min\left\{\delta_N, \frac{\delta_N \varepsilon}{M_{K_{N+1}}}\right\}$$

$\square$

Theorem. (Montel's theorem) Let $\{f_n\}$ be analytic in a domain D, which is uniformly bounded on each compact subset of D. Then $\{f_n\}$ is a subsequence which converges uniformly on all compact subsets of D.

Proof. Use the Arzela Ascoli theorem and there is a subsequence $\{f_{n_k}\}$ which converges uniformly on K. $\square$

Example. Let $D = \{\mathrm{Re}(z) > 0\}$, the right have plane. Let $f : D \to U$ be analytic with $f(1) = \alpha$. Find the upper bound of $|f'(1)|$.

Proof. Take

$$\phi = \frac{z-1}{z+1}$$

Using Schwartz's lemma for $F = \phi_\alpha \circ f \circ \phi^{-1}$, $F(0) = 0$ and

$$|F'(0)| \leq 1$$

with

$$F'(0) = \phi'_\alpha(\alpha) f'(1) \phi'(0) = \frac{1}{1-|\alpha|^2} f'(1) 2$$

Concluding,

$$|f'(1)| \leq \frac{1-|\alpha|^2}{2}$$

$\square$

Example. Let Ω be the upper halfplane $\{z \mid \mathrm{Im}(z) > 0\}$ and $f : \Omega \to U$ be an analytic function with $f(2i) = \alpha$. Find the upper bound of $|f'(2i)|$.

Proof. First find a linear fractional transformation ψ from Ω onto U satisfying $\psi(2i) = 0$. Consider

$$\psi(z) = \frac{z-2i}{z+2i}$$

For any $x \in \boldsymbol{R}$, $|\psi(x)| = 1$. If $\alpha \in \Omega$ ($\mathrm{Im}\, \alpha > 0$), then

$$\psi(z) = \frac{z-\alpha}{z-\bar{\alpha}}$$

satisfies

$$|\psi(x)| = \left| \frac{x-\alpha}{x-\bar{\alpha}} \right| = 1$$

for $x \in \boldsymbol{R}$. From this we understand that ψ is a linear fractional transformation from $\{\mathrm{Im}\, z > 0\}$ onto U. Thus, construct $F : U \to U$ that is analytic with $F(0) = 0$. We use

$$\psi = \frac{z-2i}{z+2i}$$

satisfying $\psi(\Omega) = U$, $\psi^{-1}(U) = \Omega$, and $\psi^{-1}(0) = 2i$.

$$F = \phi_\alpha \circ f \circ \psi^{-1}$$

which leads to $F(0) = 0$ and

$$F'(0) = \phi'_\alpha(\alpha) \cdot f'(2i) \cdot [\psi^{-1}]'(0)$$

Concluding,

$$|f'(2i)| \leq \frac{1-|\alpha|^2}{4}$$

$\square$